

Rapid change in coral composition alters the footprint of mass bleaching

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Highlights:

- Recurrent climate extremes have modified resilience on the Great Barrier Reef
- High-frequency bleaching events inhibit coral recovery, lowering bleaching in 2024
- Longer recovery times select for fast-growing corals, amplifying recent bleaching
- History and composition determine regional patterns of bleaching and mortality

Keywords: Climate change; biodiversity change; Mass bleaching; Resilience; Species composition; Coral Reefs

Abstract:

The Great Barrier Reef experienced five mass bleaching events in the eight years from 2016 to 2024. Each mass bleaching event can transform coral abundance and diversity, yet less is known about the long-term consequences of high bleaching frequencies and shortening recovery intervals. Using bleaching and mortality surveys in 2016 and 2024, we show that an unprecedented frequency of disturbances on the Great Barrier Reef has modified the mix of species within coral assemblages, generating a shift in how reefs respond to extreme climate events. Prohibitively high frequencies of climate extremes in far North regions (3-4 events since 2016) have delayed reef recovery and diminished bleaching and mortality levels in 2024, a pattern consistent with strong selective pressure in the region. Longer recovery intervals in central and Northern regions (7-8 years since the last major event), led to rapidly recovering assemblages with increased abundances of fast-growing, heat-sensitive species (e.g., branching and tabular *Acropora*). These regions showed unusually severe bleaching and mortality in 2024, matching or exceeding levels in 2016, and equaling declines among intact assemblages in the South in 2024 where heat stress was twice as extreme. The emerging spatial footprint of diminished recovery and shifts to heat-sensitive species reveals an expanding influence of bleaching history and composition on the resilience of the Great Barrier Reef. While reef bleaching thresholds continue to shift, rapid compositional change is creating more volatile dynamics under accumulating climate stress, including “self-amplification” where mass bleaching events favor the species most susceptible to damage.

Main Text:

Ecosystem dynamics are changing because of new disturbance regimes brought about by climate change^{1,2}. Intensification of extreme weather events such as heatwaves, storms, droughts, fires, and floods are restructuring ecosystems and giving rise to new species assemblages with limited resemblance to those studied previously³⁻⁶. Interactions between climate change and biodiversity change are important to understand future ecosystem dynamics⁷ and are becoming central in international policy frameworks to address climate and ecosystem breakdown^{8,9}. Severe or frequent climate extremes can transform biodiversity (the numbers, composition, and

abundances of species in ecosystems), leading to an increase in the prevalence of systems with a modified capacity to resist or recover from increased climate stress^{10–13}.

The increase in extreme summer sea temperatures in the past four decades represents a major turning point for coral reefs, causing widespread physiological stress and mortality in reef species^{14–16}. Recurrent marine heatwaves are impacting reefs worldwide^{17–19}, causing severe bleaching and mortality of corals at frequent intervals^{15,21–23}. Historical exposure to recurrent bleaching is altering responses of coral assemblages to subsequent bleaching events^{24–26}, driven by adaptation and acclimation within coral species^{26–29} and differences in mortality and replenishment rates among species^{21,30–33}. Taxonomic differences in bleaching due to extreme temperatures are well-documented^{34–36}. However, less is known about selective mortality, or of selective recovery in the aftermath of multiple, severe marine heatwaves, and the knock-on effects for subsequent mass bleaching. Changes in coral composition due to selective mortality can increase the relative abundances of heat-tolerant coral species and non-coral organisms (e.g., turf and fleshy algae), leaving reefs depleted but also lowering susceptibility to bleaching in subsequent years^{24,37}. In other cases, disturbances such as marine heatwaves can also favour the species most susceptible to damage³⁸, especially if there is enough time in between disturbances to allow for the recovery of heat-sensitive coral taxa with life history traits that facilitate rapid population replenishment (e.g. fast recruitment and growth)^{1,3}.

We investigated the links between compositional change and successive marine heatwaves on the Great Barrier Reef (GBR), a system impacted by five documented episodes of mass coral bleaching from 2016 to 2024^{39–41}. Early signs of bleaching on the GBR appeared intermittently in the 1970s–1990s⁴² followed by the first two mass bleaching events in 1998 and 2002⁴³. The next mass bleaching event, in 2016, drove an unprecedented, regional-scale shift in species composition^{15,44}. This was followed by four subsequent mass bleaching events in 2017, 2020, 2022, and 2024^{39–41}, with additional signs of an undocumented bleaching event in 2021 (i.e., up to six events in eight years)⁴⁵. Bleaching patterns have varied over space and time because of distinct spatial patterns of heat exposure⁴⁴ and the legacy of previous events affecting subsequent bleaching responses^{1,24}. Nevertheless, we are still learning about

interactions between compositional change and bleaching dynamics as marine heatwaves become increasingly frequent and widespread^{46,47}. Prior to 2024, the most severe Great Barrier Reef bleaching event was in 2016^{1,15}. To examine the legacy of this event and others that followed, we compared data on bleaching incidence and mortality obtained using independent surveys conducted in 2016 and 2024. These data were combined with satellite-derived data on sea surface temperatures^{48,49} to test for changes in bleaching incidence from 2016 to 2024 and relate this to differences in heat stress and coral composition. Our analysis provides insights into how rapid biodiversity change is impacting susceptibility to successive climate extremes.

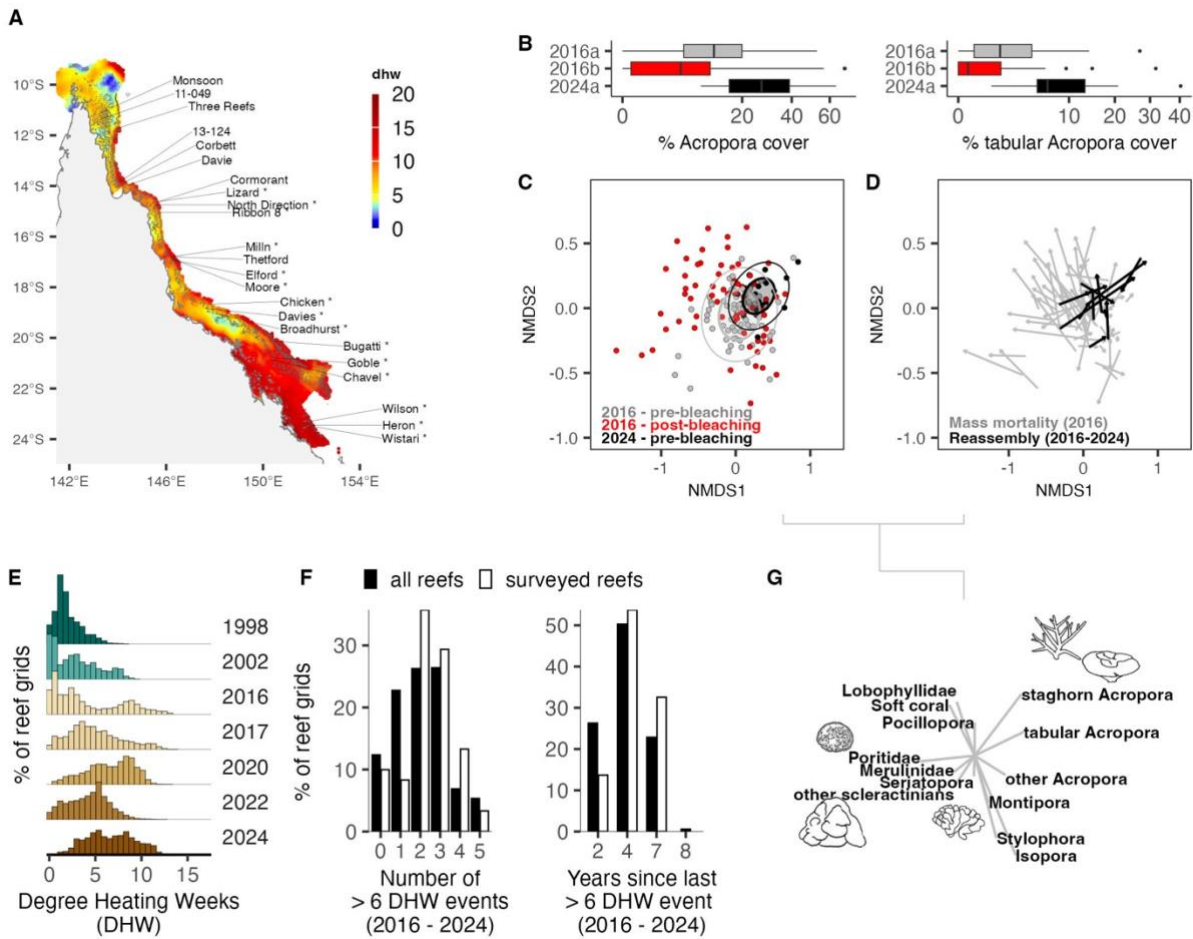


Fig. 1. Recovery and reassembly following successive heat stress events on the Great Barrier Reef. (A) Heatmap of degree heating weeks (DHW, °C-Weeks) in 2024. Reef names indicate locations of surveys conducted within 4-8 weeks of peak DHW. Reefs marked with an asterisk (*) were revisited for follow-up mortality surveys in October 2024. (B) Percent cover of *Acropora* and tabular *Acropora* across reefs at three

time points, (1) in March 2016 (pre-bleaching, n=86 reefs), (2) November 2016 (post-bleaching, n=69), and (3) March 2024 (pre-bleaching, n=23). Axes are on a square root scale. (C-D) Shifts in coral composition across three time points, indicated by non-metric multidimensional scaling (NMDS) of reefs by 13 taxa. Ellipses in (C) show 50% and 95% confidence limits for pre-bleaching NMDS scores in March 2016 and March 2024. Mass mortality of coral assemblages in (D) is shown for 69 reefs, while reassembly is presented for 10 reefs visited in 2016 and 2024. (E) Frequency plot of heat exposure ($^{\circ}\text{C}$ -Weeks) for each mass bleaching event since 1998 based on gridded satellite data from NOAA's Coral Reef Watch. (F) Histories of climate extremes across reefs in the 2016 – 2024 period, showing the percent of 25km² reef cells exposed to different numbers of $> 6^{\circ}\text{C}$ -Weeks events since 2016 and percent of reef cells with different intervals since the most recent $> 6^{\circ}\text{C}$ -Weeks event prior to 2024. (G) NMDS vectors for the plots in panels C and D.

Reassembly during recurrent climate extremes

Coral assemblages on the Great Barrier Reef have shifted towards a new mix of species following an unprecedented incidence and severity of major disturbances over 8 years (Fig. 1), ultimately leading to modified bleaching dynamics in 2024 (Fig. 2).

Comparisons of coral assemblages in 2016 and 2024 reveal widespread shifts in taxonomic composition (Fig. 1B-D), such that reefs in 2024 are occupying a distinct taxonomic space compared to reefs surveyed in 2016 (Fig. 1C). On many reefs, the percent cover of *Acropora* spp. prior to mass bleaching in 2024 exceeded the levels observed prior to the 2016 bleaching (Welch's $T = -6.91$, $p < 0.001$). This difference was particularly apparent for tabular *Acropora* (Welch's $T = -5.98$, $p < 0.001$, Fig. 1B; Fig. S1), which has high recruitment rates and fast growth⁵⁰. *Acropora* spp. experienced major declines in 2016, especially in the northern and far northern regions where heat exposure and coral mortality was severe¹⁵. In 2024, however, recovering coral assemblages across the GBR had increased percent cover of rapidly growing coral species such as *Acropora* spp. (Fig. 1D; Fig. S1), especially in northern and central regions of the GBR (Fig. S1).

Widespread compositional shifts have occurred following an increased frequency of marine heatwaves and mass bleaching events. The unprecedented scale of the 2024

109 heat stress event reinforces a trend of increasing disturbance severity across the GBR
110 (Fig. 1A, E; Fig. S2). Within the 2016-2024 interval, 39% of reef locations (25km² cells
111 encompassing reef) have experienced at least three summers of heat exposure greater
112 than 6 degree heating weeks (°C-weeks, reflecting the seasonal accumulation of heat
113 stress). Only 13% of reef locations have experienced no heat stress > 6°C-weeks, and
114 5% have experienced five heat stress events > 6°C-weeks (one for each mass bleaching
115 year since 2016, Fig. 1F). The median interval since the last event over 6°C-weeks is now
116 4 years, while 22% have experienced a 7-year gap, and only 0.5% of previously impacted
117 reefs experienced the full 8-year gap in heat exposure between 2016 and 2024 (Fig. 1F).
118 These patterns indicate that reefs are experiencing a diminishing period for recovery
119 and reassembly due to increased incidence of marine heatwaves. Indeed, while some
120 taxa that are fast growers are able to recover within ten years, the replacement of long-
121 lived corals will necessarily take many decades.

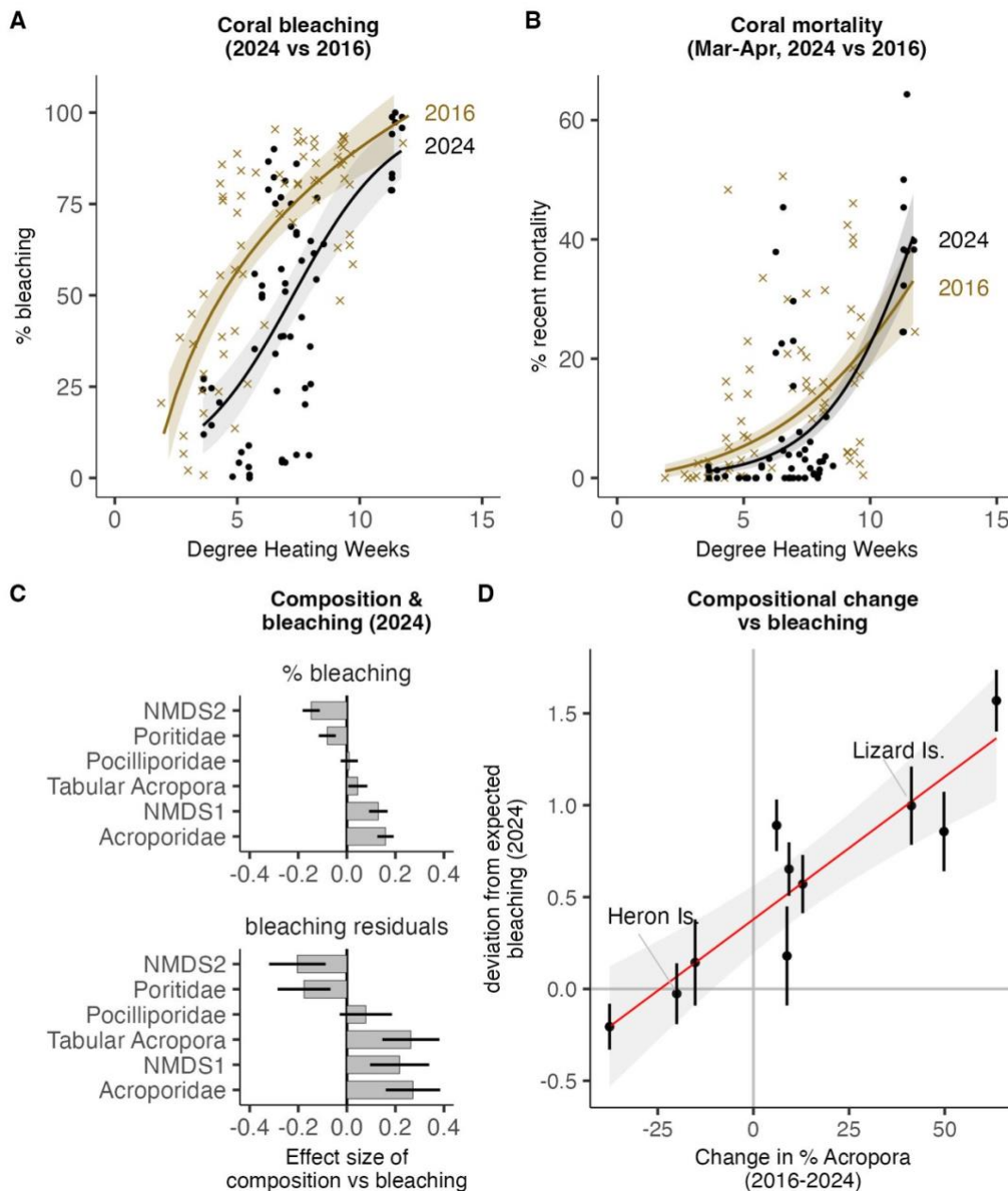


Fig. 2: The influence of heat exposure and coral composition on mass bleaching in 2024. (A) The relationship between heat exposure ($^{\circ}\text{C}$ -weeks) and bleaching, and (B) Immediate mortality (recorded during bleaching), based on surveys in March-April 2016 and March-April 2024. The grey line and crosses indicate the relationship in 2016. The relationship is fitted using beta regression (2024) and logarithmic curves (2016). (C) The effects of coral composition on bleaching patterns in 2024, indicated by standardized effect sizes and 95% confidence intervals for six compositional metrics against % bleaching (upper panel) and residuals of the DHW-bleaching relationship (lower panel). NMDS scores are based on composition in 2024 (Fig. S4). (D) Relationship (linear

regression as shown by red line) between bleaching residuals and change in percent cover of *Acropora* from November 2016 to March 2024, shown for reefs in which surveys were conducted at the same reef locations in both years.

Modified bleaching dynamics with changing coral composition

Patterns of recovery and reassembly across the GBR were strongly linked to mass bleaching patterns in 2024 and were instrumental in explaining deviations from expected bleaching for a given level of heat stress (Fig. 2). Bleaching and early-onset mortality were triggered by rapid increases in sea surface temperatures (SST) in 2024 (Fig. 2A-B), with 74% of reefs along the 2,300km length of the GBR experiencing heat stress over 4°C-weeks (Fig. 1A, E). Severe reef bleaching (>30% of coral colonies bleached) occurred at heat exposures of > 6 °C-weeks, and 75-100% bleaching was found at sites exposed to the highest levels of DHW (> 12 °C-weeks Fig. 2A). Bleaching in deeper (reef slope) habitats was 48% on average at 5-10 °C-weeks and close to 100% at 10-12°C-weeks (Fig. S3). Nevertheless, despite the strong influence of DHW on bleaching, there was substantial variation in bleaching severity among reefs experiencing similar levels of heat stress (Fig. 2A), particularly between 5 to 10 °C-weeks (mid-levels during summer 2024). Coral composition and history influenced whether reefs showed higher or lower bleaching than expected for a given DHW exposure (Fig. 2C-D).

Unusually high bleaching occurred on reefs with high abundances of heat-susceptible taxa which have increased in abundance on many reefs in the 2016-2024 interval (Fig. 2C-D). Deviations from expected bleaching were estimated from the residuals of the best fit model for the DHW-bleaching relationship (a beta regression), and residuals from a General Additive Model (GAM) were also analysed to check the robustness of the results (table S1). Reefs with the highest levels of bleaching in 2024 for a given amount of heat exposure were typically characterised by high abundances of heat-sensitive staghorn and tabular *Acropora*, and lower abundances of massive and branching *Porites* (Fig. 2C; Fig. S4). Moreover, reefs that have recouped high abundance of heat-susceptible *Acropora* since 2016 showed elevated bleaching severity compared to what would be expected for their DHW exposure in 2024 (linear fit, $p < 0.001$ and $R^2 = 0.87$, Fig. 2D). For example, while diverse coral assemblages on Heron Island bleached

severely due to extremely high DHW levels, Lizard Island reefs also bleached severely despite lower DHW exposure because of a dominance of *Acropora*, which had regrown following severe disturbances⁵¹ (Fig. 2D). These localised and region-wide patterns indicate that recovering reefs with increased abundances of heat-susceptible taxa are more susceptible to severe mass bleaching.

Colony-level bleaching severity for a given DHW level was generally lower in 2024 compared to 2016 ($P < 0.001$, Fig. 2A; Fig. S5). However, twenty percent of reef sites surveyed in 2024 bleached above the average predicted level in 2016 (points above the grey curve, Fig 2A), and comparisons with aerial scores suggest that bleaching in 2024 was also more severe than 2017 and 2020 at higher levels of heat exposure (Fig. S5). In general, the 2024 relationship between DHW and bleaching reveals a stronger distinction between heavily bleached and unbleached reefs relative to 2016, particularly at mid-levels of heat stress. The large disparity in bleaching between reefs experiencing similar levels of heat stress, in addition to lower curve fits in 2024 (adjusted $R^2 = 0.50$, $P < 0.001$ and adjusted $R^2 = 0.39$, $P < 0.001$ for GAMs from 2016 and 2024, respectively), suggests that our capacity to predict in-water bleaching from DHW values alone is diminished. Instead, reefs are showing increased variability in their bleaching responses due in part to shifts in composition.

Large numbers of corals died rapidly within 4-8 weeks of the onset of bleaching in 2024 (Fig. 2B), including in recovering coral assemblages and especially on reefs exposed to extreme DHW levels. Recently dead colonies in March 2024 accounted for up to 43% of coral colonies at some sites (averaged across transects) and reached 86% on individual transects, indicating that large numbers of colonies had already succumbed to high heat stress within a few weeks of peak DHW exposure (Fig. 2B; S3). Immediate coral mortality recorded during bleaching (4-8 weeks after peak DHW) were lower in 2024 compared to 2016 (Fig. 2B), yet substantial mortality still occurred at moderate heat exposure. Most of the initial survivors were severely bleached at the time of censusing and remained at risk of further mortality over the coming months⁵².

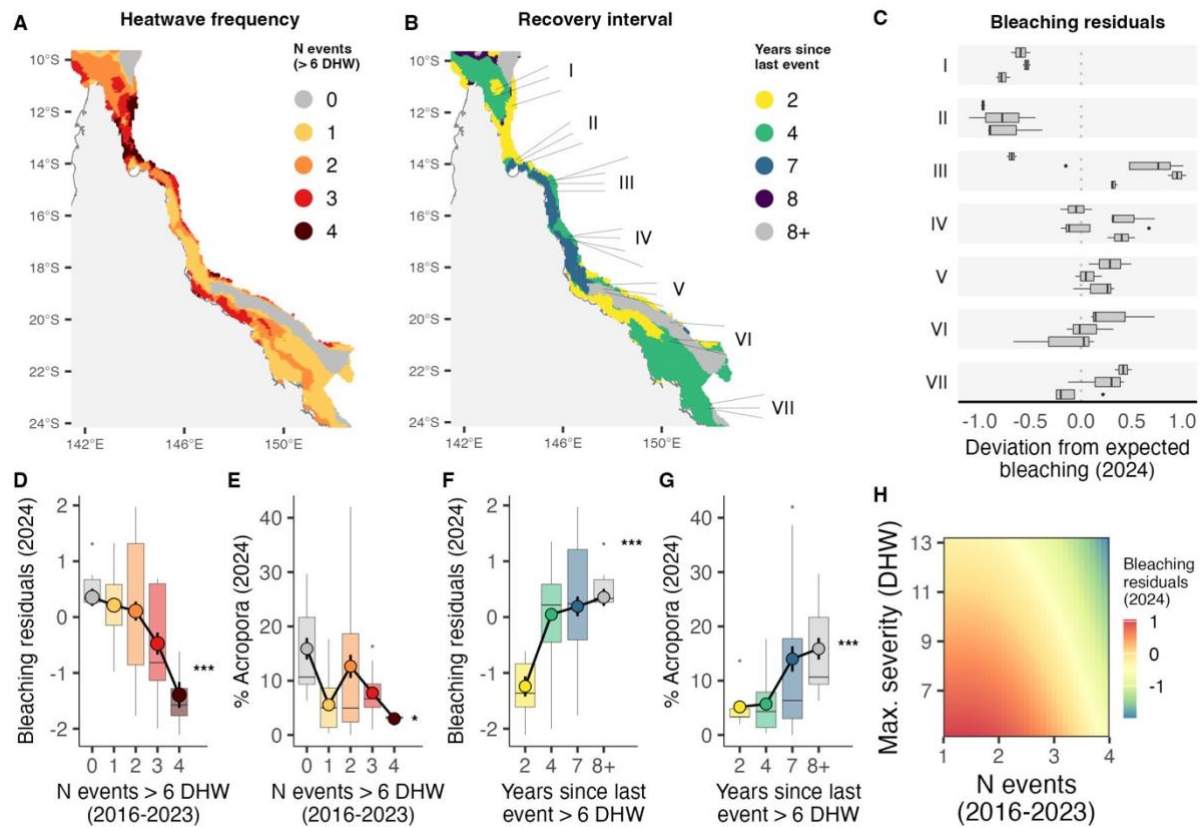


Fig. 3. The influence of disturbance and recovery interval on coral composition and bleaching susceptibility. (A). Spatial patterns of the frequency of > 6 DHW events across 25km² grid cells from 2016-2023. (B) Spatial patterns of the time interval (in years) since the last > 6 DHW event across 25km² grid cells from 2016-2023. (C) Bleaching residuals in 2024 at the reef-level across 7 reef regions. (D-E) The effects of the frequency of > 6°C-weeks events from 2016-2023 on bleaching residuals and percent *Acropora* abundance in 2024. (F-G) The effects of the interval since the last > 6°C-weeks event on bleaching residuals and percent *Acropora* abundance in 2024. In (D-G), means and standard errors are presented with trendline, and the significance relationship is represented using “*” (* p < 0.05, ***p < 0.001). (H) Heatmap depicting the influence of heat stress frequency (number of events > 6°C-weeks) and heat stress severity (maximum DHW experienced) on bleaching residuals in 2024.

Regional histories of disturbance and recovery

Differences in composition and their resulting effects on bleaching severity can be attributed to the recent history of large-scale disturbance and recovery across regions of the Great Barrier Reef (Fig. 3; Fig. S2). Since 2016, the individual reefs surveyed in this study have experienced between 0 and 4 severe bleaching events prior to 2024, and the time since the preceding event before 2024 varied among reefs from 2 years (2022-2024) to 8 or more years (since the 2016 mass bleaching, or earlier in 2002 and 1998). The highest frequencies of heat stress events ($> 6^{\circ}\text{C}$ -weeks) were found in far Northern regions (e.g., Princess Charlotte Bay and Cape Grenville), where large areas experienced $> 6^{\circ}\text{C}$ -weeks in 2016, 2017, 2020, and 2022. The longest intervals prior to 2024 since the last event $> 6^{\circ}\text{C}$ -weeks (excluding reefs with no exposure in the 2016-2024 interval) were found in Central and Northern Regions (Fig. 3A-B). While many reefs within these regions experienced milder bleaching in 2020 and 2022, declines in coral cover were limited in these years⁵³, whereas these reefs were severely damaged in 2016 and 2017¹⁵. These patterns reveal highly distinct histories of heat exposure across the Great Barrier Reef, including widely disparate recovery intervals among reefs that have been affected by past heat stress events.

Histories of climate stress influenced spatial patterns in both coral composition and bleaching incidence relative to DHW in 2024 (Fig. 3D-G; Fig. S6). The lowest levels of *Acropora* abundance in 2024 were found on reefs with the highest incidence of heat stress events since 2016 and therefore the shortest intervals for recovery (Fig. 3E, G), typically in the far North (e.g., Princess Charlotte Bay and Cape Grenville). Moreover, these reefs had low bleaching relative to DHW exposure, despite all surveyed reefs experiencing $> 4^{\circ}\text{C}$ -weeks, suggesting that compositional change has subdued bleaching in these locations (Fig. 3D, F). The highest abundances of *Acropora* in 2024 were found on reefs with low heatwave frequencies and long recovery times (1-2 events $> 6^{\circ}\text{C}$ -weeks within 8 years, concentrated in Central and Northern Regions), and these reefs typically had higher bleaching than expected relative to DHW, suggesting that compositional change has elevated bleaching in these locations (Fig. 3D-G). The relationship between heat stress history and recent bleaching was found in both shallow and deep habitats and was consistent when using residuals from the GAM and

beta-regression (Fig. S7). Bleaching patterns and abundance of heat susceptible species in 2024 were therefore significantly aligned with histories of heat stress, with fewer heating events favouring higher abundances of *Acropora* and higher levels of bleaching in 2024.

Our results support demographic forecasts indicating that bleaching every 2 years (equivalent to 3 or more bleaching events within the 2016-2024 interval) does not give heat-susceptible populations with fast growth and high rates of recruitment (such as *Acropora*) enough time to recover, thereby leading to diminishing abundances of these species over time³. The effects of heat stress frequency (more directly linked to accumulating mortality) on reducing *Acropora* abundance were most pronounced when reefs were exposed to more than 2 events (Fig. 3E; Fig. S7). While reefs with > 7 years since the last heat stress event (more directly linked to recovery times) had higher abundances of *Acropora*, this effect was most pronounced for tabular *Acropora* (Fig. S9), a group that tends to recover rapidly after disturbance because of high rates of growth and recruitment⁵⁴ (primarily in shallow reef crest habitats, Fig. S7). Consequently, the relationship between composition and recent climate history (Fig. 3E, G) supports the expectation that when recovery intervals are sufficiently long, reefs can recoup high abundances of rapidly recovering species, yet longer recovery windows can also trigger higher levels of bleaching due to a higher abundance of heat-susceptible taxa.

Mass bleaching and mortality

Subsequent reef surveys in October 2024 revealed substantial mortality of corals in 2024, which are largely attributable to the extreme heat stress and mass bleaching recorded earlier (in February) that year. Coral mortality for a given bleaching level (% colonies bleached) was lower in 2024 compared to 2016 (Figure 4a). Mortality was also lower for a given DHW in 2024 compared to 2016 (Figure 4b), a pattern potentially attributed to rapid adaptation and acclimation within coral species^{26–29}. The principal driver of spatial variation in mortality in 2024 was the % cover of *Acropora* (Figure 4c) and the interval since the last heatwave event (Figure 4d), suggesting that coral composition and heatwave history now has a major influence on levels of bleaching-induced mortality⁵⁵. Notably, the maximum recorded cover of *Acropora* spp in 2024 was

higher than in 2016 (Figure 4c), and it was the minority of sites with highest cover of *Acropora* spp. that exhibited the greatest coral loss. On this subset of reefs, coral assemblages have not become more resistant to bleaching-induced mortality despite potential adaptation over eight years, largely due to a shift in coral composition.

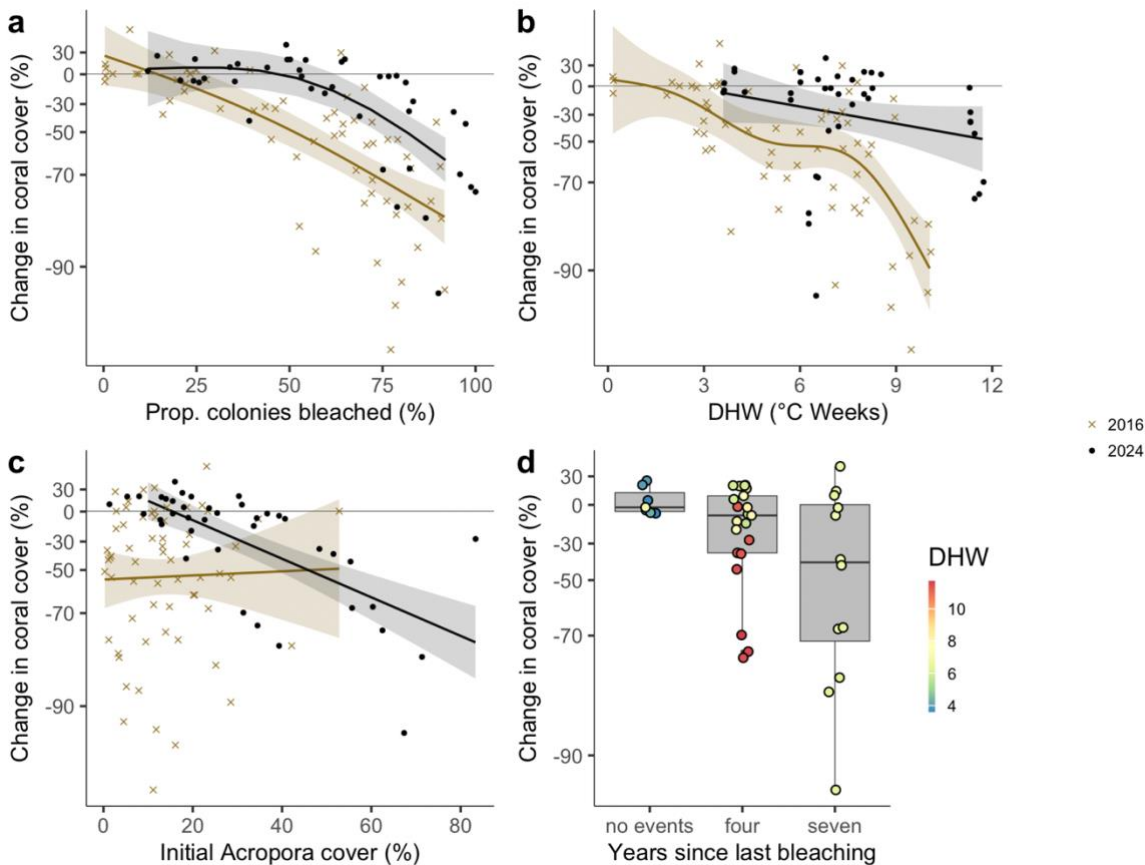


Figure 4: Annual bleaching-driven mortality of corals in 2016 versus 2024, reflecting changes in coral cover from April to October in each year; (a) Relationship between % colonies bleached versus long-term coral mortality. (b) Relationship between reef DHW versus long-term coral mortality. (c) Relationship between initial (April) *Acropora* abundance versus long-term coral mortality. (d) Changes in coral cover in 2024 across reefs with different intervals since the last heatwave event.

Conclusions

Mass bleaching and mortality on coral reefs due to anthropogenic heating is increasingly dependent on the spatial mosaic of recovery and reassembly in the aftermath of previous events⁵⁶. This study shows that shifts in coral assemblages are altering ecosystem trajectories and climate-susceptibility under repeated bleaching.

Most reefs in 2024 had lower than expected levels of bleaching compared to 2016, a pattern corroborated by aerial surveys during previous events in 2017, 2020 and 2022^{1,24}. Increased thermal tolerance and reduced bleaching incidence are largely a consequence of the selective filtering of heat-susceptible coral taxa under unprecedented frequencies of mass bleaching (Fig. 2-3) and by the selective removal of heat-susceptible phenotypes within coral populations²⁶⁻²⁹. Nevertheless, we also identified recovering reefs with high abundances of heat-susceptible taxa that were more likely to undergo severe bleaching and subsequent mortality. These reefs experienced fewer bleaching events in recent years, creating a recovery window which allowed some fast-growing taxa to return (Fig. 3). Ultimately, regions of the Great Barrier Reef are experiencing too many bleaching events to recoup high coral abundances, and too short a recovery interval to return to mature reef assemblages. Today, this dynamic is unfolding on virtually all reefs as they experience shortening recovery intervals between successive climate extremes^{17,18,57,58}.

Our study highlights suggests that shifts in coral composition and are interacting with climate impacts to affect ecosystem dynamics, especially with increasing incidence of marine heatwaves and mass coral bleaching. Compositional changes are an overwhelming component of global biodiversity change⁵⁹, with shifts in the local mix of species highly prevalent in ecosystems exposed to increasing climate stress². In other systems (e.g., forests), interactions between different disturbances (e.g., fires, wind, disease) are often found to have amplifying effects on ecological change, while composition change has dampening effects (e.g., limiting the impact of subsequent disturbances)^{2,60}. The shifts in coral composition found in this study are both dampening and amplifying reef ecological change, contingent on recent bleaching history. Despite potential for adaptation over eight years, recovering reefs were still highly susceptible to severe heat-driven mortality in 2024 owing to a shift in composition. The boom-and-bust dynamics generated by these changes are likely to have dramatic consequences for future trajectories of coral cover and corresponding ecosystem function⁶¹, possibly increasing some processes (e.g., biomass growth) while diminishing others (e.g., biomass storage). In contrast, intact and mature assemblages

313 supporting diverse functions are likely to become increasingly rare with recurrent
314 sequences of climate-driven mass mortality and reassembly.

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Materials and Methods

Bleaching surveys: Surveys of reef assemblages were conducted in March–April 2024 across 23 reefs in 7 distinct regions (Cape Grenville, Princess Charlotte Bay, Lizard Island, Cairns, Cape Bowling Green, Hydrographers Passage, and the Capricorn Bunker Group) along Australia’s Great Barrier Reef (GBR). At each reef, we conducted 3 replicate 50m point intercept transects and 3 replicate 10 × 1m belt transects on the

reef crest (1-3m depth) and reef slope (6-9m depth) at each of the 2-3 sites. Transects were laid at a constant depth and parallel to the reef contour. Point transects were conducted to quantify the percent cover and composition of corals and other benthic organisms, and included 48 coral groups including tabular *Acropora*, staghorn *Acropora* and other *Acropora* (typically corymbose or digitate taxa), alongside 15 other benthic groups including soft corals, crustose coralline algae, and macroalgae. 100 uniformly spaced points (50cm apart) were sampled per transect. Belt transects were conducted to quantify bleaching and recent mortality, with each coral colony mostly or wholly contained within the 10 × 1m transect area identified within the 48 coral groups and given a bleaching score between 1 and 8 (1 being no bleaching, 5 being fully bleached, and 8 being recently dead). The total number of colonies within bleaching categories 2-8 were used to calculate percent bleaching for each reef. Categories 6-8 (partially or fully dead) were used to calculate percent recent mortality. Juvenile corals (< 5 cm diameter) were surveyed but excluded from current analyses.

Heat stress data: We extracted 25km² resolution sea surface temperature (SST) and degree heating week (DHW) data based on site-level GPS readings from the National Oceanic and Atmospheric Administration's database via Coral Reef Watch ^{48,49}. Maximum DHW across the summer months at weekly intervals was used to measure heat exposures throughout the GBR in the 2016-2024 interval (focussing on 2016, 2017, 2020, 2022, and 2024; Fig S1). The frequency of heat stress events was determined by the number of times DHW levels exceeded 6 DHW from 2016 to 2024, which is historically enough to trigger widespread mortality among diverse coral species¹⁵. Time since the last heat stress event was estimated by the time since the most recent year that heat exposure exceeded 6 DHW prior to 2024.

DHW-bleaching relationship: A range of models were used to test the association between heat exposure and the severity of bleaching (Fig. 2a; Table S1) and were conducted separately for shallow and deep sites (Fig. S4). Models of the relationship included linear, polynomial, logarithmic (i.e., saturating), exponential, and logistic (i.e., sigmoidal) regression, beta regression and general additive model (GAM), as presented in Table S1. While the GAM had the best model fit (lowest AIC value), contrasts to results from previous studies⁴⁴ were necessarily based on non-GAM relationships. The

best-fitting non-GAM model (i.e., the beta regression, Fig. 2a) revealed significant variation in bleaching incidence relative to reef-level heat stress between 2016 and 2024. Subsequent analyses used both the GAM and beta regression to test the robustness of our results against model selection decisions. The 2016 curve fit was measured using logarithmic regression following ref.⁴⁴, which generates the equation for the fitted line of $y = 48.6\ln(x) - 21.6$. Following previous analyses based on aerial surveys²⁴, we also analysed the influence of DHW on the probability of severe bleaching (> 30% corals bleached) using a generalized linear model with a binomial error structure (Fig. S6).

Coral composition: We analysed coral composition in two ways. First, we conducted a non-metric multidimensional scaling analysis (NMDS) using 15 coral groups, with the remaining 48 taxonomic groups, which independently accounted for much lower levels of percent cover amalgamated into an “other coral” category. The coordinates of the NMDS (from axes NMDS1 and NMDS2) were then used as an indicator of coral composition (Fig. 2c). We also used simple metrics of composition indicated by the total or relative percent cover of three the three predominant coral families, Acroporidae, Pocilloporidae (including *Pocillopora*, *Stylophora*, and *Seriatopora*) and Poritidae (including branching and massive taxa). The average site-level percent cover of individual taxa were used in analyses of coral composition (Fig. 2e). Tabular *Acropora* were also analysed separately in some cases because of the tendency for tabular morphotypes to dominate on partially recovered reefs^{3,54}. To examine compositional change, we pooled 15 coral groups into 12 taxa to align with 2016 surveys and conducted an NMDS of reefs at three time points: (1) In March 2016 (pre-bleaching, n=86 reefs), (2) In November 2016 (post-bleaching, n=69), and (3) In March 2024 (pre-bleaching, n=23).

Bleaching incidence relative to DHW: Our analysis quantifies unexpectedly high or low susceptibility to bleaching as the deviation from the relationship between DHW and bleaching (Fig. 2a), which were quantified from the model residuals separately for shallow and deep habitats (Fig. S4). Residuals were extracted from the beta regression and GAM describing the DHW-bleaching relationship to test for a possible influence of model selection on our results (with little effect found, see Fig. S8). Our analyses

subsequently linked bleaching susceptibility to various metrics of reef condition and location, including GBR region, composition, bleaching history, and coral recovery. Linear relationships between site-level compositional metrics and (1) total bleaching and (2) bleaching residuals were used to quantify the influence of reef composition on reef bleaching responses. Composition metrics were scaled (Z-transformed) prior to analysis and slope and 95% confidence interval were used to quantify effect sizes (i.e., standardised regression coefficients, Fig. 2e). We used ANOVA to quantify the link between the frequency and interval between heat stress events, bleaching susceptibility, and taxon abundance to test for possible associations between recent climate history and reef bleaching. Finally, links between bleaching susceptibility and recent recovery were estimated for reefs that were assessed in both 2016 and 2024 ($n = 10$), with the 2016 surveys conducted in October following mass bleaching and mortality¹⁵. The percent change in *Acropora* abundance was calculated based on differences in transect surveys between the two years. Change in *Acropora* abundance was subsequently linked to bleaching susceptibility using a best-fit linear relationship (Fig. 2d).

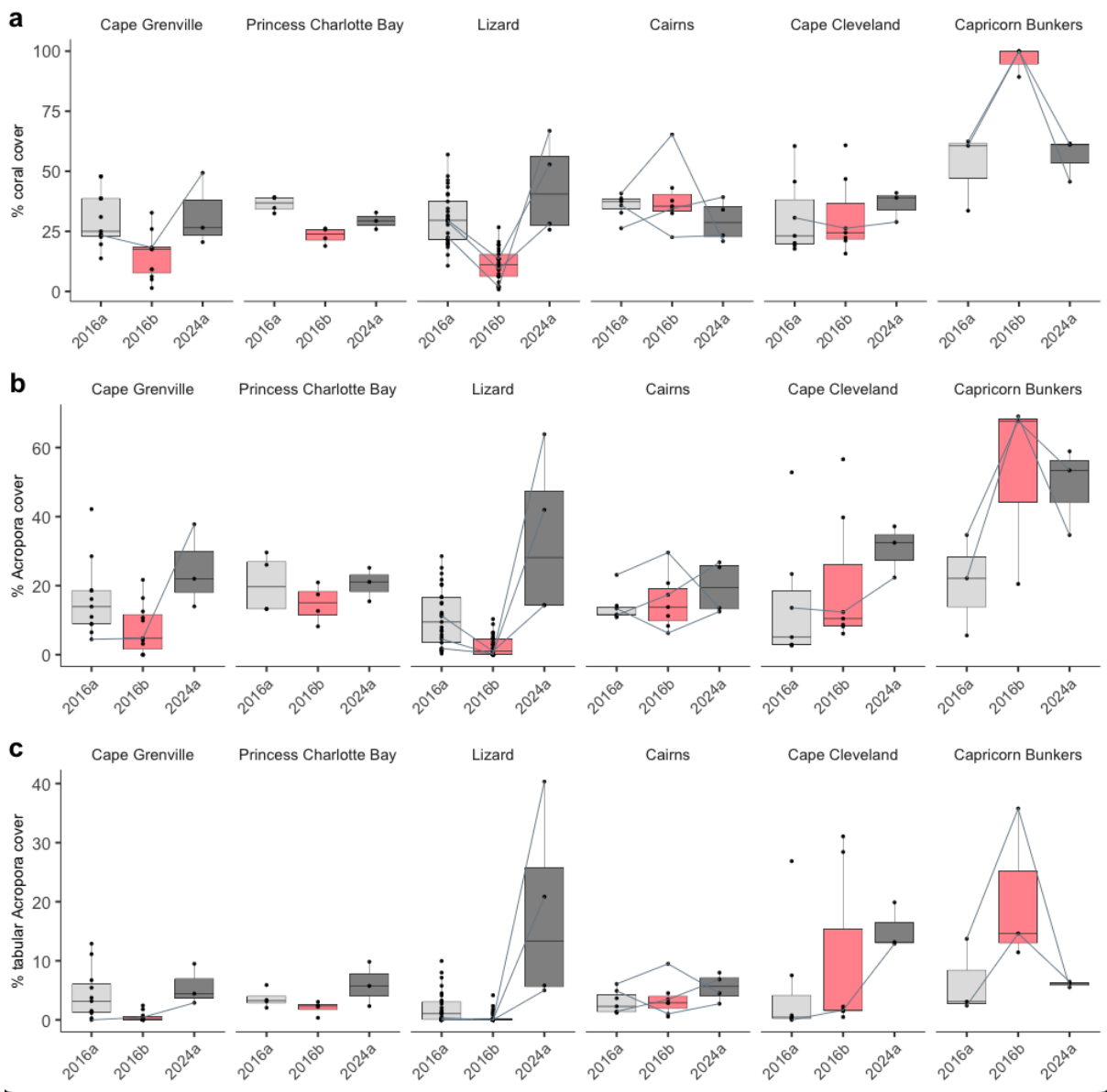
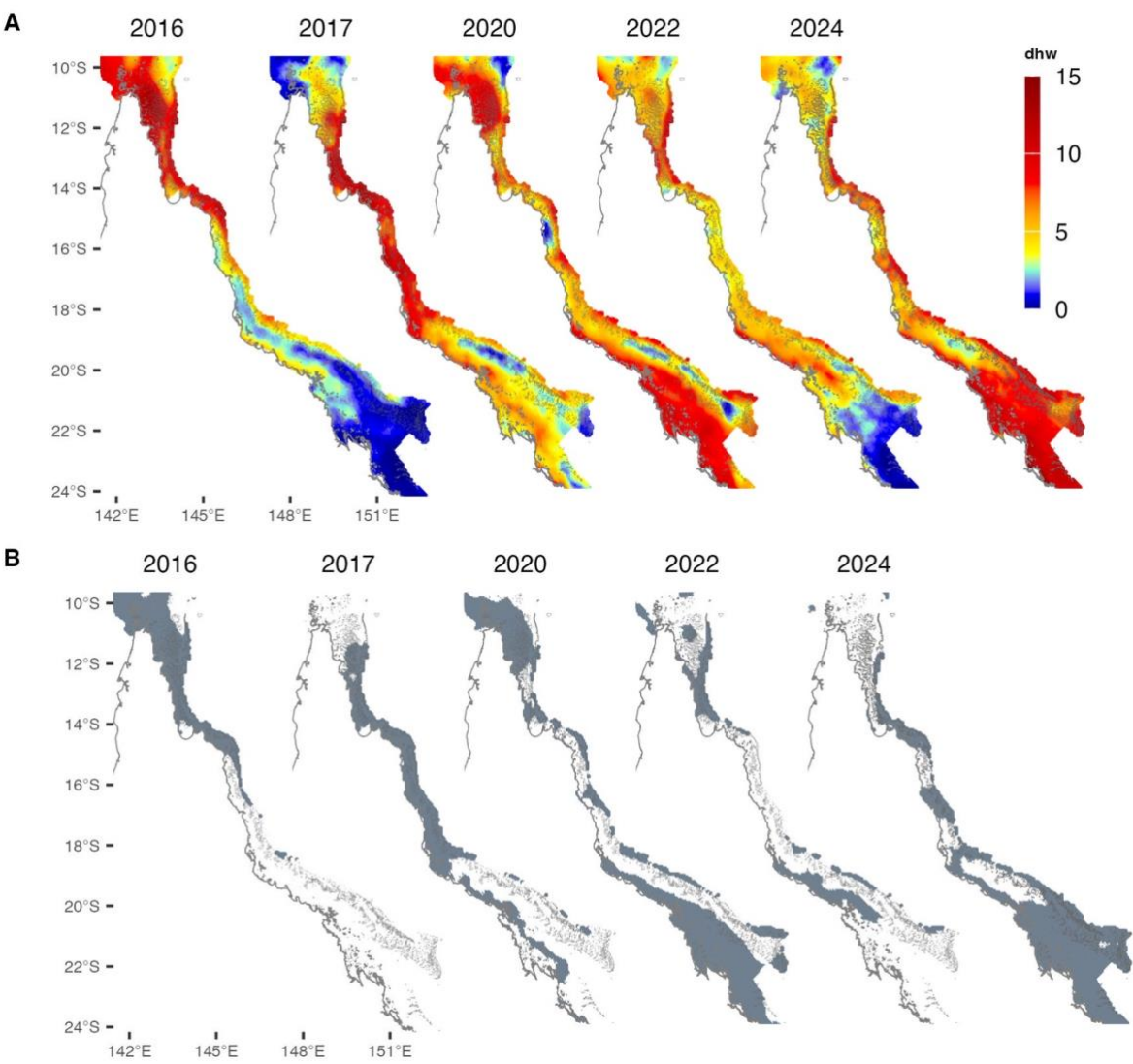


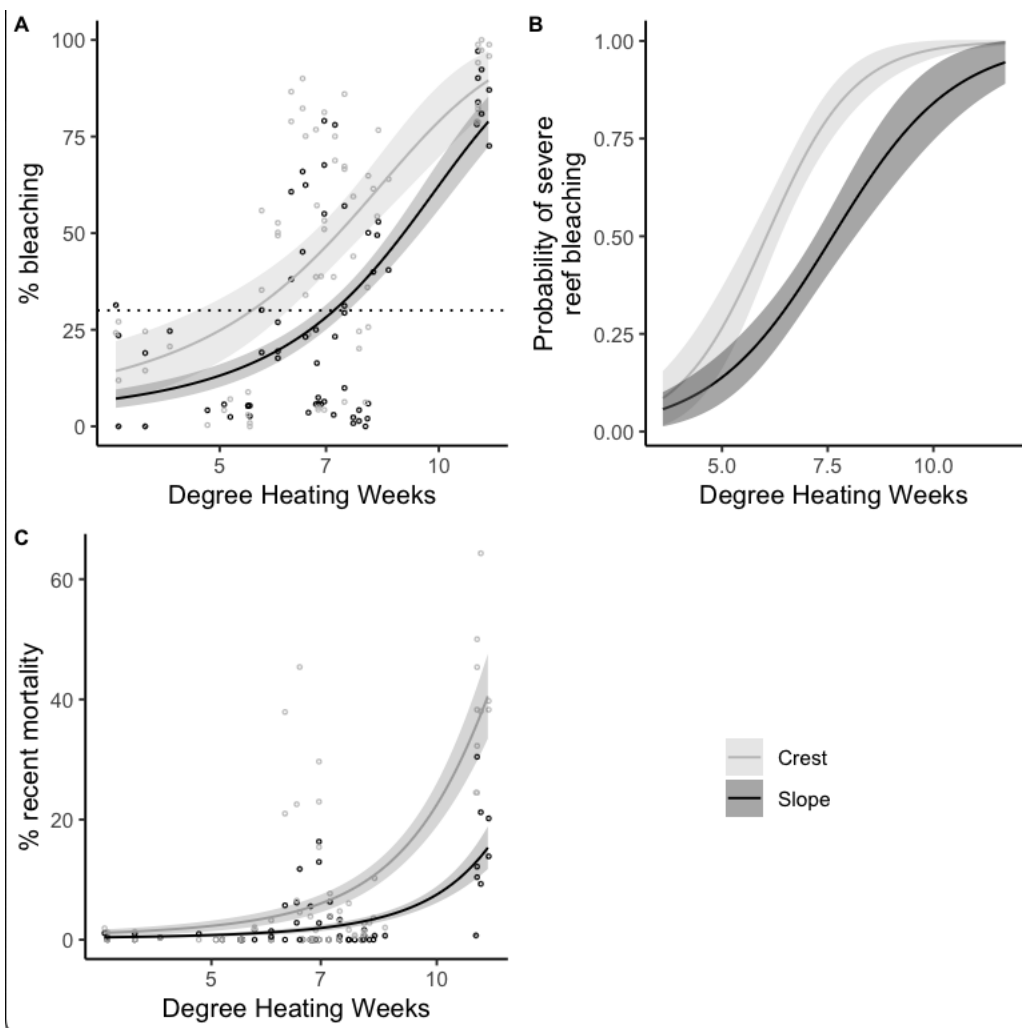
Fig. S1. Change in percent coral cover (a), percent *Acropora* cover (b) and tabular *Acropora* cover (c), across all reefs visited along the length of the GBR at three time points: (i) 2016 before bleaching-induced mortality, (ii) 2016 after bleaching-induced mortality, and (iii) 2024 before bleaching-induced mortality. Box plots for each metric at each time interval are shown. Reefs revisited in each year are linked by a line. Rows are shown for each GBR region from North to South, excluding Hydrographers Passage for which there was insufficient data from 2016.



630 **Fig. S2.** (A) Heatmaps of maximum degree heating weeks (DHW) in 2016, 2017, 2020,
631 2022, and 2024. These data were used to estimate frequency of heat stress events and
632 recovery intervals, presented in Fig. 3a-b. (B) Spatial distribution of reefs that
633 experienced > 6 DHW in 2016, 2017, 2020, 2022, and 2024 (shown in grey). The
634 underlying data were used to calculate frequency of heat stress events in Fig. 3a and
635 time since last heat stress event in Fig. 3b.

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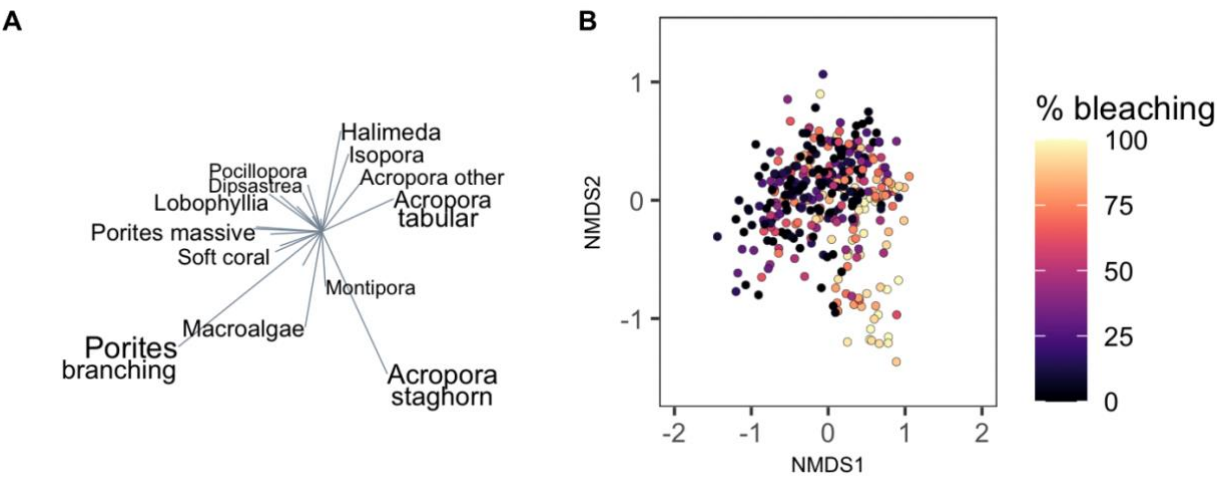
638

639 **Fig. S3.** Patterns of bleaching and mortality in two habitats, shallow crests (1-3m depth)
640 and slopes (6-9m depth) in 2024. (A) Heat exposure versus percent of colonies
641 bleached. The cut-off for severe reef bleaching (30%) is shown by the dotted line. (B)
642 Heat exposure versus probability of severe reef bleaching for each habitat. Curves show
643 a generalized linear model with binomial error structure. (C) Heat exposure versus
644 percent recent coral colony mortality in March-April 2024. Curves in (A) and (C) are
645 based on a beta regression.

646

SUPPLEMENTARY MATERIAL

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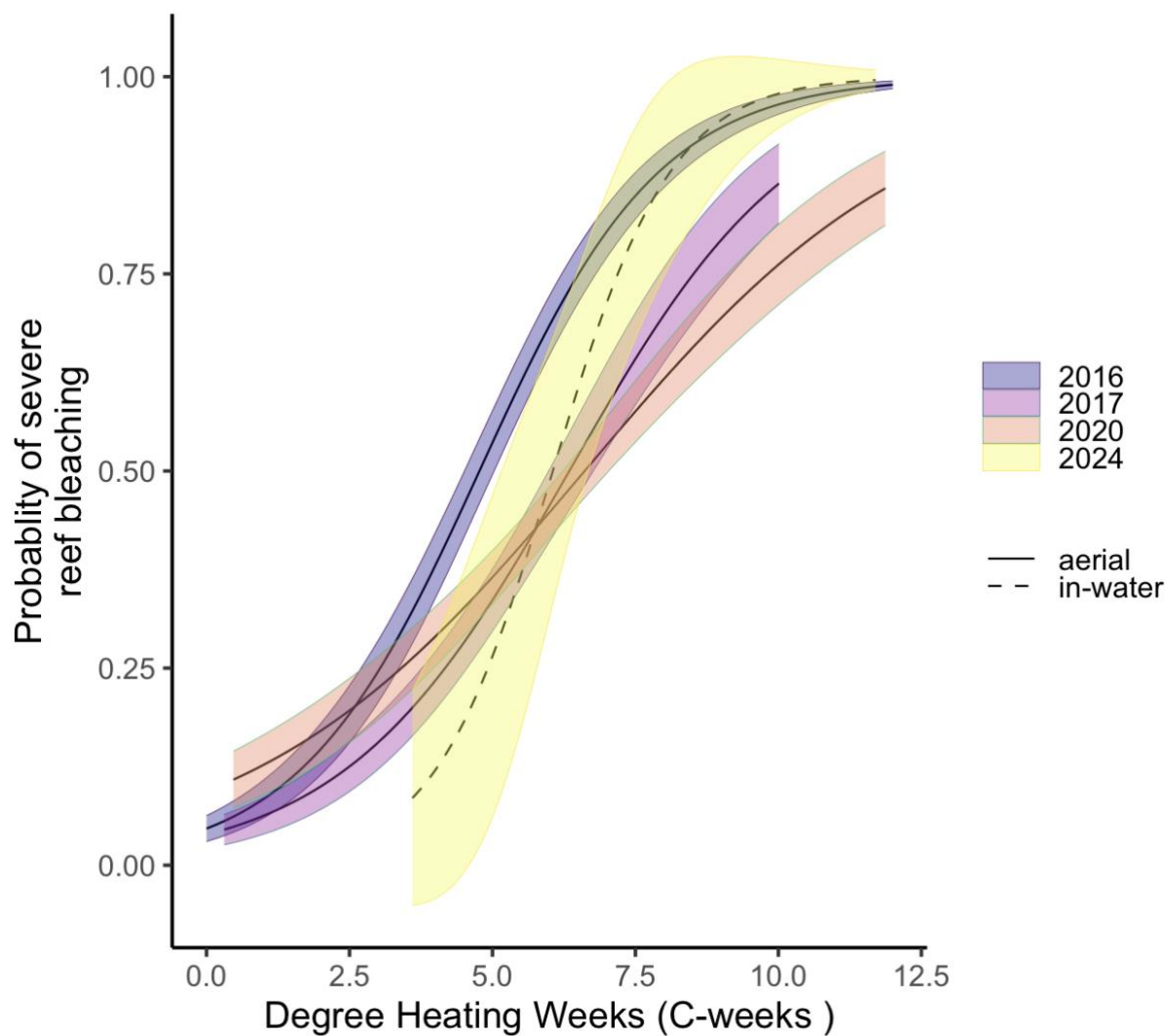
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649 **Fig. S4.** Non-metric multidimensional scaling analysis of each reef site by benthic taxa,
650 coloured by total percent bleaching. The analysis is used to quantify the effects of
651 multidimensional composition in 2024 on bleaching in 2024 (see Fig. 2c). Some taxon
652 labels in (A) are omitted for clarity.

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SUPPLEMENTARY MATERIAL



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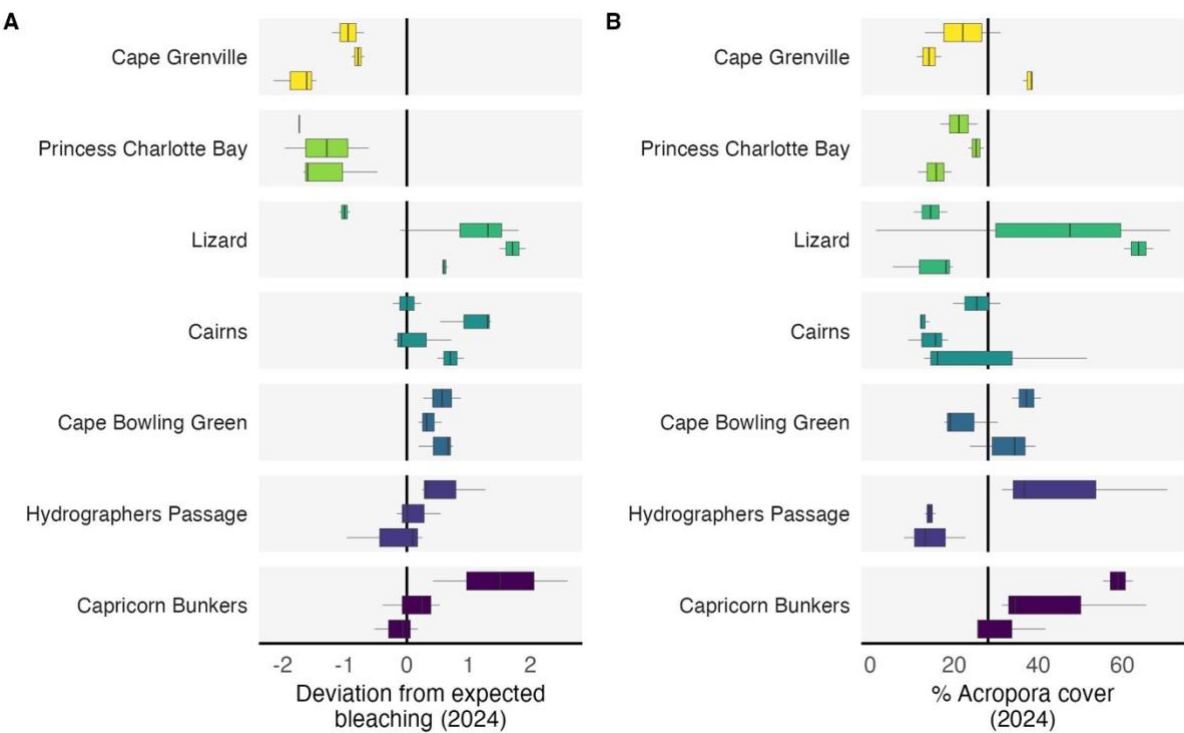
656

657 **Fig. S5.** Heat exposure versus chance of severe reef bleaching, comparing in-water
658 bleaching surveys (2024) with aerial scores from previous years (2016, 2017, 2020)
659 derived from Hughes et al 2021. Note that the 2024 surveys differ in methodology and
660 sample size, yet both use the same cut-off of 30% colonies bleached to define severe
661 bleaching. Curves are derived from a generalized linear model with binomial error
662 structure.

663

SUPPLEMENTARY MATERIAL

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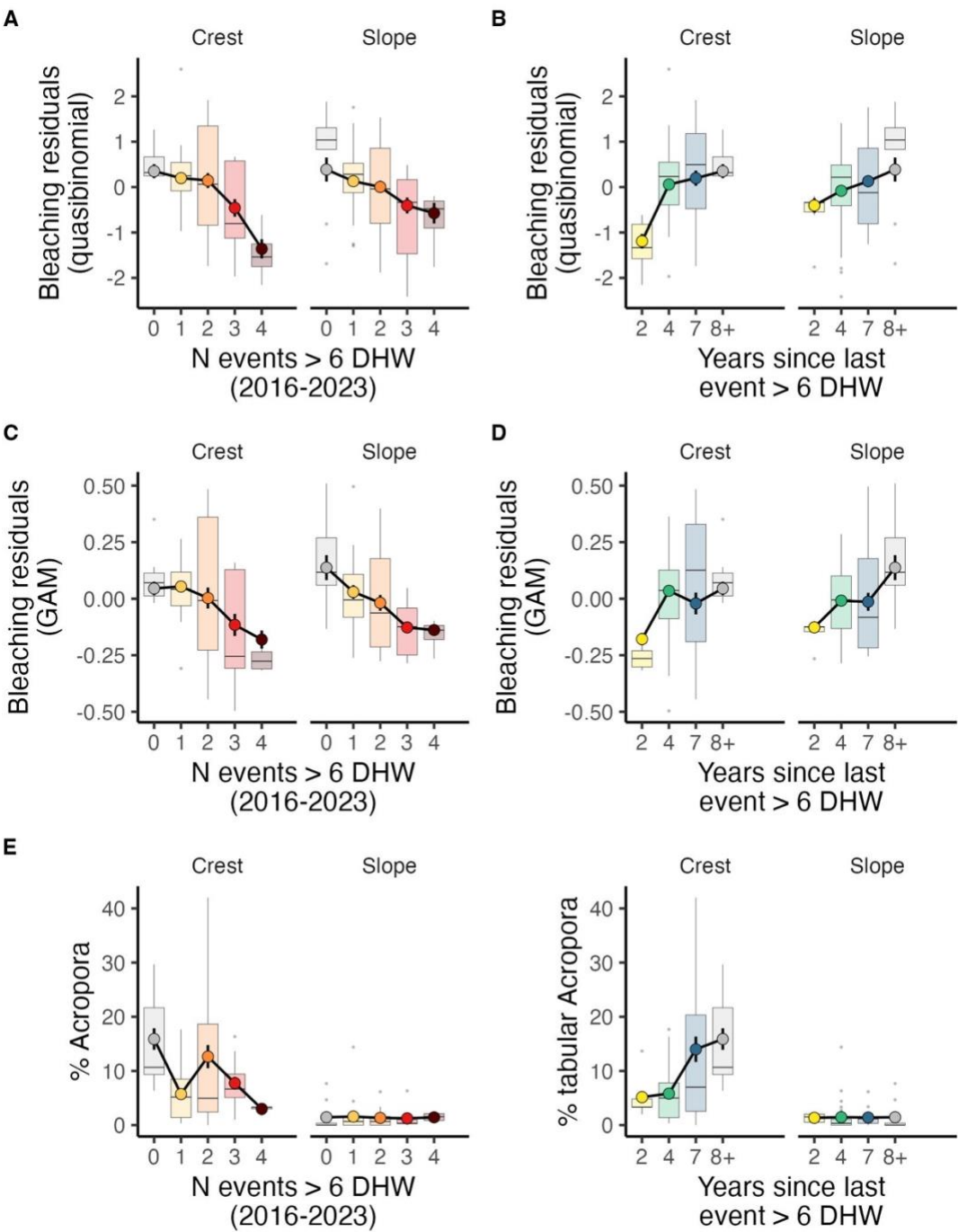
665

666 **Fig. S6.** Geographic patterns in (a) deviations from expected bleaching, as determined
667 by the residuals of the DHW-bleaching relationship, and (b) percent cover of *Acropora*
668 spp. Each boxplot represents a reef, coloured by region and listed from North to South.
669 Vertical lines indicate zero (a) and the total mean percent cover of *Acropora* (b).

670

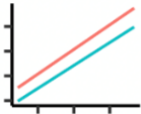
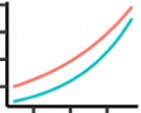
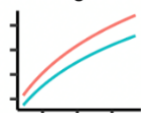
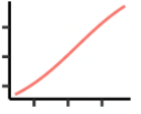
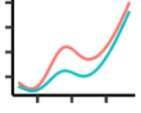
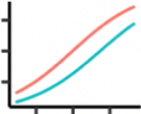
SUPPLEMENTARY MATERIAL

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672

673 **Fig. S7.** Influence of heat stress frequency (a) and time since last heat stress event (b)
674 on bleaching susceptibility and reef composition. Patterns are shown for two habitats
675 (shallow crests and deep slopes). In each panel, points and trend lines show the mean
676 and standard error. Boxplots show the distribution of data.

Model	R ²	AIC	Fit
Linear	Crest: 0.41 Slope: 0.41	Crest: 5.25 (df = 3) Slope: -2.92 (df = 3)	
Exponential	Crest: 0.24 Slope: 0.20	Crest: 6.78 (df = 3) Slope: -9.32 (df = 3)	
Logarithmic	Crest: 0.40 Slope: 0.37	Crest: 7.39 (df = 3) Slope: 4.30 (df = 3)	
Logistic	NA	Crest: 7.70 (df = 4) Slope: NA	
General additive model	Crest: 0.42 Slope: 0.48	Crest: 5.25 (df = 3) Slope: -8.39 (df = 5.5)	
Beta regression	NA	Crest: - 6.72 (df = 4) Slope: -15.7 (df = 4)	

679 **Table S1.** Models used to quantify relationships between heat exposure (DHW) and
680 bleaching and subsequent residuals (i.e., deviations from expected bleaching). Plot
681 examples (last column) show the fitted relationships between DHW (x-axis) and percent
682 bleaching (y-axis). Bold lettering shows the highest values for each column.